

Overview and Function:

Folate is a water-soluble B-vitamin (vitamin B9) that is naturally present in foods. Folic acid is the synthetic form of folate used in fortified foods and most dietary supplements. Folate functions as a coenzyme in the synthesis of nucleic acids (DNA and RNA) and in the metabolism of amino acids, including the conversion of homocysteine to methionine.

Recommended Dietary Allowances (RDA) in Dietary Folate Equivalents (DFE):

The FNB established the following RDAs:

- Children 1-3 years: 150 mcg DFE/day
- Children 4-8 years: 200 mcg DFE/day
- Children 9-13 years: 300 mcg DFE/day
- Adolescents 14-18 years: 400 mcg DFE/day
- Adults 19+ years: 400 mcg DFE/day
- Pregnant Females: 600 mcg DFE/day
- Lactating Females: 500 mcg DFE/day

Neural Tube Defect Prevention:

Adequate folate intake is critical prior to conception and during early pregnancy. Periconceptional intake of 400 to 800 mcg of folic acid daily substantially reduces the risk of neural tube defects, specifically spina bifida and anencephaly.

Dietary Sources of Folate:

Folate is found naturally in a wide variety of plant foods:

- Dark Green Leafy Vegetables: Spinach, kale, Brussels sprouts, and asparagus.
- Legumes and Nuts: Black-eyed peas, kidney beans, peanuts, and sunflower seeds.
- Fruits: Oranges, orange juice, papayas, and bananas.
- Fortified Foods: In the United States, enriched grains (flour, bread, rice, pasta, and cornmeal) are mandated to be fortified with folic acid (140 mcg per 100 g).

Tolerable Upper Intake Level (UL):

The UL for folate applies ONLY to synthetic folic acid from fortified foods and supplements, not natural food folate:

- Children 1-3 years: 300 mcg DFE/day
- Children 4-8 years: 400 mcg DFE/day
- Children 9-13 years: 600 mcg DFE/day
- Adolescents 14-18 years: 800 mcg DFE/day
- Adults 19+ years: 1,000 mcg DFE/day

High intakes of folic acid can mask the hematologic signs of vitamin B12 deficiency while allowing neurological damage to progress.